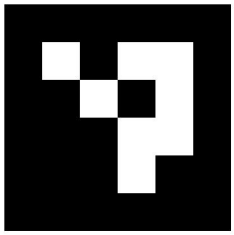
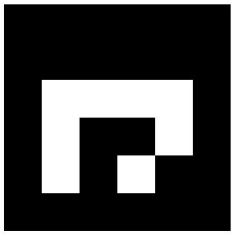
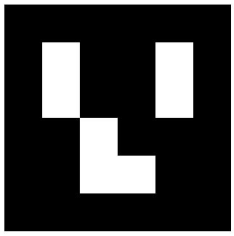
ID 0



ID 1



half-size breadboard here



ID 3



ID 2

Polarity marker mat

1. Print at 100% / "Actual size" (no fit-to-page). The bar below must measure exactly 1 cm.

2. Tape the mat flat on the bench and keep the breadboard inside the dashed rectangle.

3. Mount the camera 25-35 cm above, looking straight down, so all four markers are visible.

Markers: ArUco DICT_4X4_50, IDs 0 (top-left), 1 (top-right), 2 (bottom-right), 3 (bottom-left), 30 mm each.

Safety: batteries and USB power only. Never mains. If anything gets hot, smells or smokes: switch off, call the TA.



1 cm scale bar: check with a ruler